

"The concept of converting mountains to towers and light rays to internet connectivity for remote areas is something to share with people who are working with electronics and technology.

"When we suggested mountain-tops as towers, normal telecom companies asked how these would be powered and who would maintain them? Since Li-Fi requires very little power, we made power systems that can be carried and are training an army of shepherds in Li-Fi rather than acquiring professionals from other places."

—Sonam Wangchuk

power required for each tower is up to 7kW. Optical units on the other hand require only 48W.

Just like the solar-heated building, Navtech, HIAL, and SECMOL have developed a solar-heated battery storage device. When the temperatures in the barrel (below the solar panel) were measured, a difference of approx. 61 degrees was obtained. Sonam says, "The temperatures in the night are a minimum of +26 degrees, even while it is a chilling -40 degrees outside. This proved the power system can work unmanned on mountain-tops."

As shown in Fig. 4, using Li-Fi receivers and transmitters, any signal (example, telecom) can be obtained from a normal tower. When a laser-emitting device is installed on the top of the mountain, any remote village below the neighbouring mountains will have internet connectivity, enabling students to have online classes. "Using this solution, a lot of cost can be saved," opines Sonam.

The devices used in these innovations are all made in India, in accordance with *Atmanirbhar Bharat*, with the battery and power system being made in the campus itself. Laser beams have a range of around 10km, but in the thin air at high altitudes the range increases up to 15km (still under testing).

Considering this to be a good distance from one mountain to



Fig. 6: Mountain-to-mountain internet connectivity



Fig. 7: Solar-heated battery storage device

another, it can also be relayed as an optical fibre cable and used for radio frequency for the valleys below where there is population. The capacity of the laser beam transmission (up to 1GB) will replace digging, power, and expenditure for the hilly areas. The speed remains the same in both optical fibre and laser beam communication.

Considering weather conditions, even though there is rain/snow on rare days, the autonomy of the power system is such that even on cloudy days, photovoltaics would generate sufficient electricity (may be only 30 to 40% of their peak), which would help get through the situation.

This technology is now finding its way with major telecom provid-

ers and being used for mountain regions. Hence using Li-Fi receivers and transmitters, signals can be obtained from a normal tower to connect remote villages. Telecom providers need to look into this technology that can help solve many problems.

"Using this solution, a lot of cost can be saved," says Sonam. In conclusion, he shares, "These are some examples of just to say that indigenisation and localisation of making things in India is truly a proud moment for the country." **EFY**

This article is based on the speech given recently by Sonam Wangchuk at the 35th Year VLSI Conference. The article has been transcribed and curated by EFY's Sharon Abhignya Katta, who is passionate about innovation, especially in the field of telecommunications.